

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
ENVIRONMENT
MODULE 2
BIOGEOCHEMICAL CYCLES

NUTRIENTS

Nutrients are the elements and compounds that organisms need to live, grow, and reproduce. **Nutrition** refers to the interrelated steps by which a living organism assimilates food and uses it for growth, reproduction and regulation of body functions.

Standing state: The amount of nutrients present in the soil at any given time is referred to as the standing state.

Nutrients can be **essential** or can have some **beneficial** role in **plant growth**. There are some essential nutrients for plants; they are known as essential nutrients because: (1) plants are unable to complete their life cycle in the absence of the mineral element; (2) the function of the element cannot be replaced by another mineral element; and (3) the element is directly involved in plant metabolism.

We can classify these nutrients into **macronutrients** and **micronutrients**. When eliminating any one of these nutrients, plants will display abnormalities of growth, deficiency symptoms, or may not reproduce normally.

Type	Definition	Examples
Macronutrients	Nutrients required by plants in large amounts	Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Phosphorus (P), Potassium (K), Calcium (Ca), Magnesium (Mg) and Sulfur (S)
Micronutrients	Elements required in trace amounts. Silicon (Si) and Cobalt (Co) are beneficial nutrients - not deemed essential for all plants but may be essential for some.	Boron (B), Chlorine (Cl), Copper (Cu), Iron (Fe), Manganese (Mn), Sodium (Na), Zinc (Zn), Molybdenum (Mo), Nickel (Ni), Silicon (Si) and Cobalt (Co)

BIOGEOCHEMICAL CYCLES

The movement of nutrient elements through the various components of an ecosystem is called **nutrient cycle** or **bio-geochemical cycle**. It is the continuous flow of elements and compounds between organisms and the land (lithosphere), air (atmosphere) and water (hydrosphere).

Term	Definition	Example
Reservoir	A place or region or location where a biogeochemical element is in its highest concentration. Elements being cycled are held and stored for some time in reservoirs.	Coal or fossil fuels are reservoirs for carbon.
Exchange Pools	When the elements are held for only short periods of time, they are being held in exchange pools.	Plants and animals, which temporarily use elements in their systems and release them back into the air or surrounding medium.

Depending upon the source, biogeochemical cycles are divided into gaseous and sedimentary cycles.

Gaseous: reservoir is atmosphere or hydrosphere Eg : C, N

Sedimentary: reservoir is Earth's crust Eg: P,S.

Another classification: Perfect and Imperfect cycles

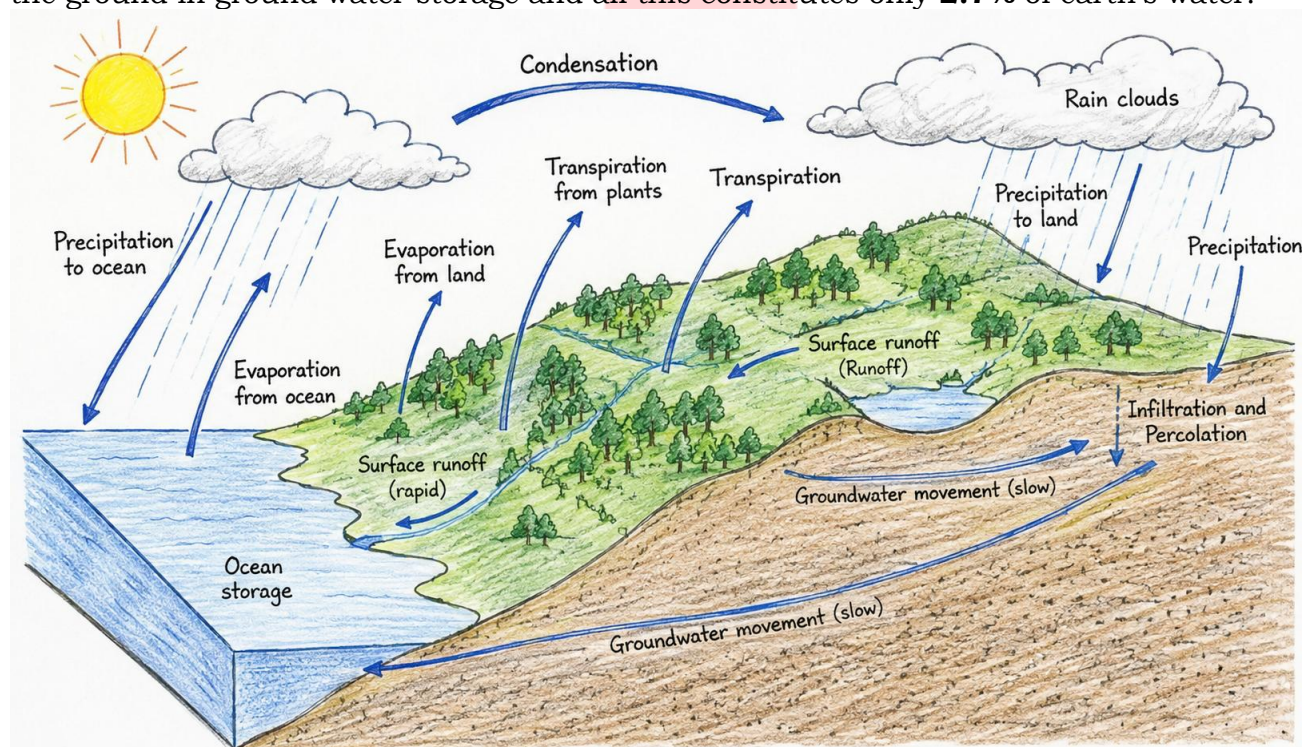
Perfect - nutrients are replaced as fast as they are utilized. Eg: Most of the gaseous cycles.

Imperfect- some nutrients are locked in the sediments & become unavailable for immediate cycling.

Eg: sedimentary cycles.

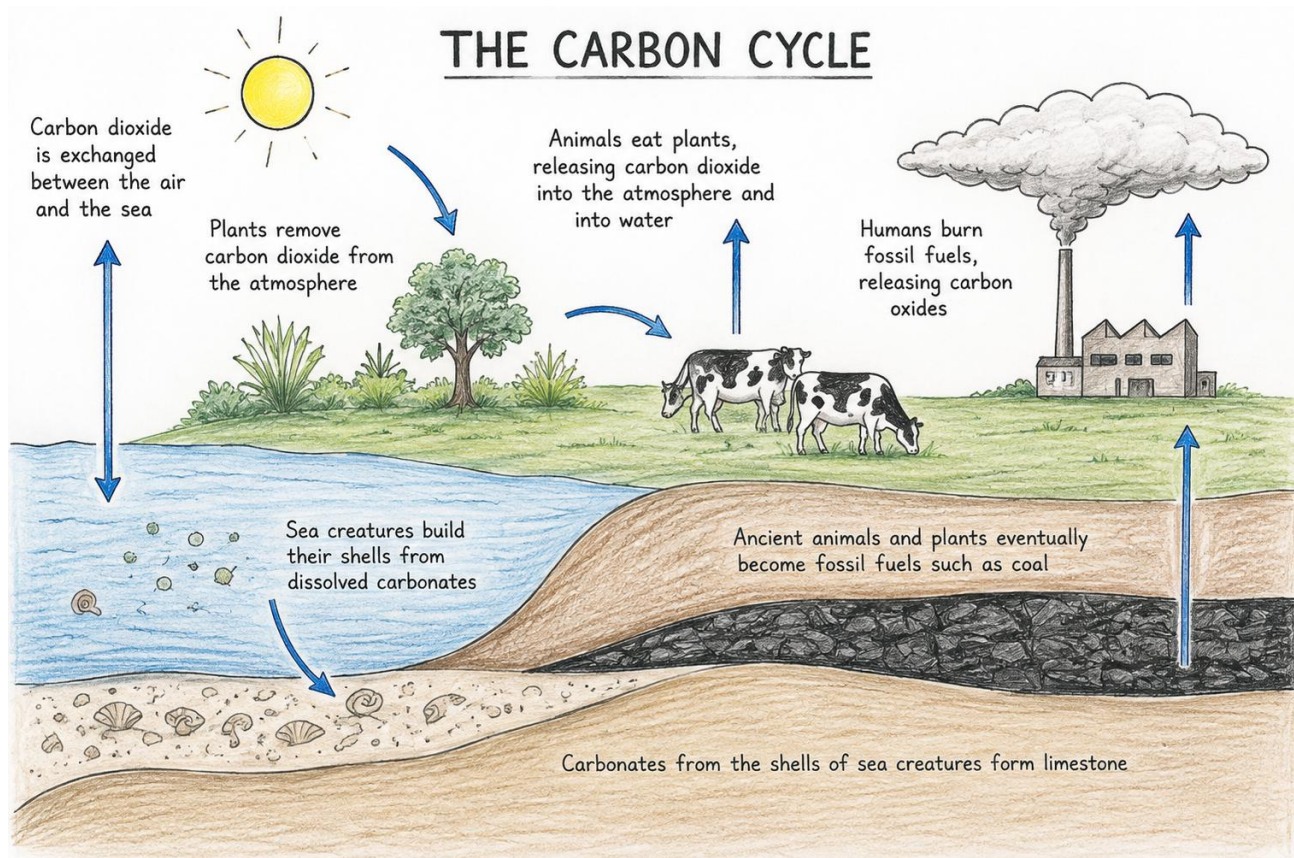
WATER CYCLE

The process by which water continually changes its form and circulates between oceans, atmosphere and land is known as the water cycle. This cycle makes water available to living things on land for its various uses. Around **97.3%** of the earth's water is stored in the ocean. The forms of freshwater available to us are stored in lakes, glaciers, snow caps, rivers, and below the ground in ground water storage and all this constitutes only **2.7%** of earth's water.



Process	Definition
Evaporation	Process by which water changes from liquid state to gas or vapour state.
Transpiration	Release of water from the leaves of plants.
Condensation	Process by which gas phase changes into liquid phase.
Precipitation	Any product of the condensation of atmospheric water vapour that falls under gravity.
Surface Run off	On land, much precipitation runs along the surface of the ground until it enters a river or stream that carries the runoff back to an ocean or lake.
Infiltration	Some precipitation seeps into the soil, deeply enough to become ground water. Water in the soil enters plants through the roots, and the water cycle begins again.

CARBON CYCLE



Carbon is a **building block** of both plant and animal tissues. Living tissue contains carbon because they contain organic molecules like proteins, lipids, nucleic acids and carbohydrates. The second composition of living organisms constitutes carbon and the first is Oxygen.

In the **atmosphere**, carbon is present as carbon dioxide gas, CO_2 . Carbon dioxide is **released** into the atmosphere by volcanic activity, aerobic respiration, decomposition of organic matter, and human activities like burning wood, exhausts of vehicles, burning fossil fuels such as coal, oil and natural gas to produce heat and energy.

Carbon **enters the biotic world** through the action of **autotrophs or producers**, they absorb CO_2 to make food (carbohydrates) by **photosynthesis** and release O_2 to the atmosphere. The carbohydrates are passed along food webs to animals and other consumers. Thus, plants help in regulating and monitoring the percentage of Oxygen and Carbon dioxide in the earth's atmosphere.

Both plants and animals release CO_2 during respiration; they also return fixed carbon to the soil in the waste they excrete and when the living organisms die. This is done by the **decomposers** in the food chain through breakdown of the dead bodies. Some of this carbon in the dead material is turned into **fossil fuels**, which is used again for meeting energy needs by combustion. These processes complete the Carbon cycle.

Carbon dioxide is a greenhouse gas contributing to global warming and climate change. Every year billions of tons of Carbon dioxide is released to the atmosphere by anthropogenic sources. Rapid deforestation, and massive burning of fossil fuels for energy and transport have significantly increased the rate of release of carbon dioxide into the atmosphere.

Ocean as a Carbon Sink

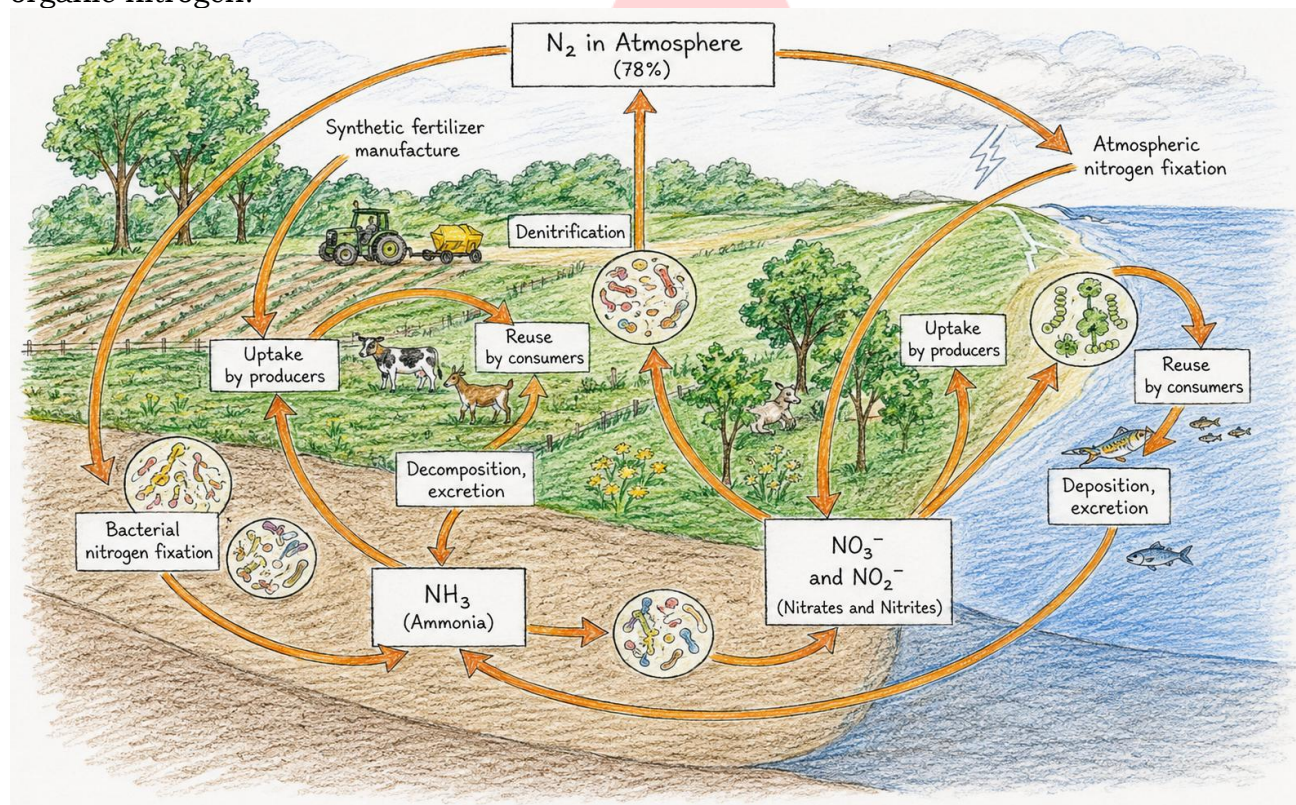
Oceans contain earth's **largest store** of **carbon**. Carbon dioxide once dissolved in seawater can remain as is or can be converted into carbonate (CO_3^{2-}) or bicarbonate (HCO_3^-). **Certain forms of sea life biologically fix** carbonates with calcium (Ca^{2+}) to produce **calcium carbonate** (CaCO_3). This substance is used to produce shells and other body parts by organisms such as corals, clams, oysters, some protozoa and some algae. When these organisms die, their shells and body parts sink to the ocean floor where they accumulate as carbonate-rich deposits. After long periods of time, these deposits are physically and chemically altered into **sedimentary rocks**.

The **producers** in aquatic ecosystems (**Phytoplankton**) growth depends on the availability of

CO_2 , sunlight and nutrients. They prepare food through photosynthesis. Phytoplanktons are the "**biological pump**" that transfers carbon from the ocean surface to the deep ocean and throughout the food chain. Carbon is stored in the lithosphere in both inorganic and organic forms. **Inorganic deposits** of carbon in the lithosphere include fossil fuels like coal, oil and natural gas, oil shale, and carbonate-based sedimentary deposits like limestone. **Organic** forms of carbon in the lithosphere include litter, organic matter and humic substances found in soils.

NITROGEN CYCLE

Nitrogen forms about **78%** of the air on earth. But **plants cannot use nitrogen directly from the air**. Nitrogen is often the most limiting nutrient for plant growth. This problem occurs because most plants can only take up Nitrogen in two forms: **ammonium ion** (NH_4^+) and the **nitrate ion** (NO_3^-). Nitrogen itself is unreactive so cannot be used by green plants to make organic nitrogen.



Nitrogen Fixation

N_2 can be converted to a usable form, nitrates. Before that it has to convert to ammonia. The process of conversion of nitrogen (N_2) to ammonia is termed '**Nitrogen-fixation**'. In nature, lightning and ultraviolet radiation provide enough energy to convert nitrogen to nitrogen oxides. Industrial combustions, forest fires, automobile exhausts and power-generating stations are also sources of atmospheric nitrogen oxides. Nitrogen is introduced to the soil from the atmosphere by precipitation (rain). Chemical industries are also producing ammonia fertilizers.

Some prokaryotic species are capable of fixing nitrogen by the use of the enzyme **nitrogenase**. Such microbes are called **Nitrogen fixers**: eg: Azotobacter, Beijernickia, Rhodospirillum, Bacillus, Rhizobium, Frankia, and some cyanobacteria such as **Anabaena** and Nostoc.

Conversion of nitrogen to ammonia by living organisms is called '**Biological Nitrogen Fixation**'. Members of the bean family (legumes) and some other kinds of plants form mutualistic symbiotic relationships with Nitrogen-fixing bacteria, the legume-bacteria relationship. Species of rod-shaped Rhizobium has such relationships with the roots of several legumes such as alfalfa, sweet clover, sweet pea, lentils, garden pea, broad bean, clover beans, etc. These plants have small outgrowths of the roots called **nodules**, in which the bacteria lives. In exchange for some nitrogen, the bacteria receive carbohydrates from the plant's carbohydrates and special structures (nodules) in roots where they can exist in a moist environment. Ammonia is used less by plants for uptake because in large concentrations it is extremely toxic; it has to be converted to nitrates.

Nitrification

Nitrates can be obtained by the conversion of ammonia by nitrifying bacteria in the soil. Ammonia is first oxidized to **nitrite** by the bacteria **Nitrosomonas** and/or **Nitrococcus**. The nitrite is further oxidized to **nitrate** with the help of the bacterium **Nitrobacter**. These steps are called '**Nitrification**'.

Ammonification and Denitrification

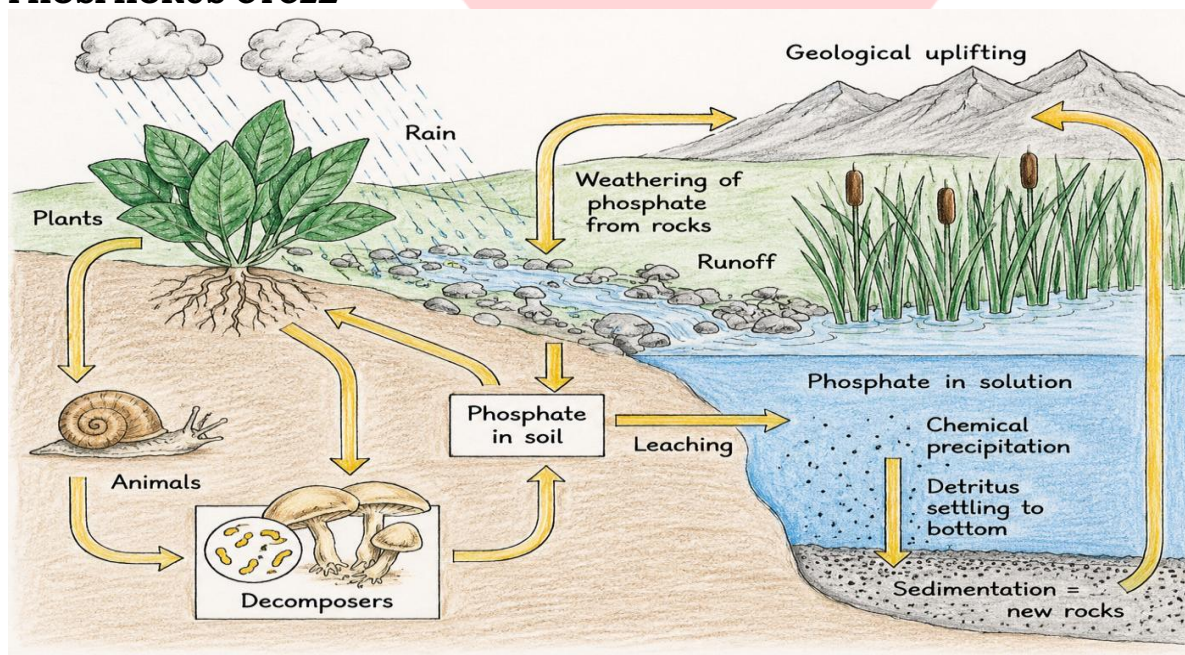
The nitrate formed is absorbed by plants and transported to the leaves. In leaves, it is reduced to form ammonia that finally forms the amino group of amino acids and is also used to produce other organic molecules. Decomposition of organic nitrogen of dead plants and animals into ammonia is called '**Ammonification**'.

However, nitrate is very soluble and it is easily lost from the soil system by leaching. Some of this leached nitrate flows through the hydrologic system until it reaches the oceans where it can be returned to the atmosphere by **denitrification** and diffuse into the atmosphere. Plants build up proteins using nitrates absorbed from the soil. When animals eat these plants, they in turn use it to build animal protein. When these animals poop, pee or die, the urea, excreta or carcass are broken down by decomposers and the nitrogen is reintroduced into the soil in the form of ammonia by ammonification. Nitrate present in the soil is broken down to nitrogen and it is sent back to air again, by the process of '**denitrification**' carried by the denitrifying bacteria **Pseudomonas** and **Thiobacillus**.

Key Bacteria in the Nitrogen Cycle

Process	Bacteria	Reaction
Nitrogen Fixation (free-living)	Azotobacter, Beijernickia, Rhodospirillum, Bacillus, Frankia, Anabaena, Nostoc	$N_2 \rightarrow NH_3$ (ammonia)
Symbiotic Fixation	Rhizobium (with legumes)	$N_2 \rightarrow NH_3$ in root nodules
Nitrification (Step 1)	Nitrosomonas, Nitrococcus	$NH_3 \rightarrow NO_2^-$ (nitrite)
Nitrification (Step 2)	Nitrobacter	$NO_2^- \rightarrow NO_3^-$ (nitrate)
Denitrification	Pseudomonas, Thiobacillus	$NO_3^- \rightarrow N_2$ (returns to atmosphere)
Ammonification	Decomposers (bacteria and fungi)	Organic N $\rightarrow$ NH_3 (ammonia)

PHOSPHORUS CYCLE

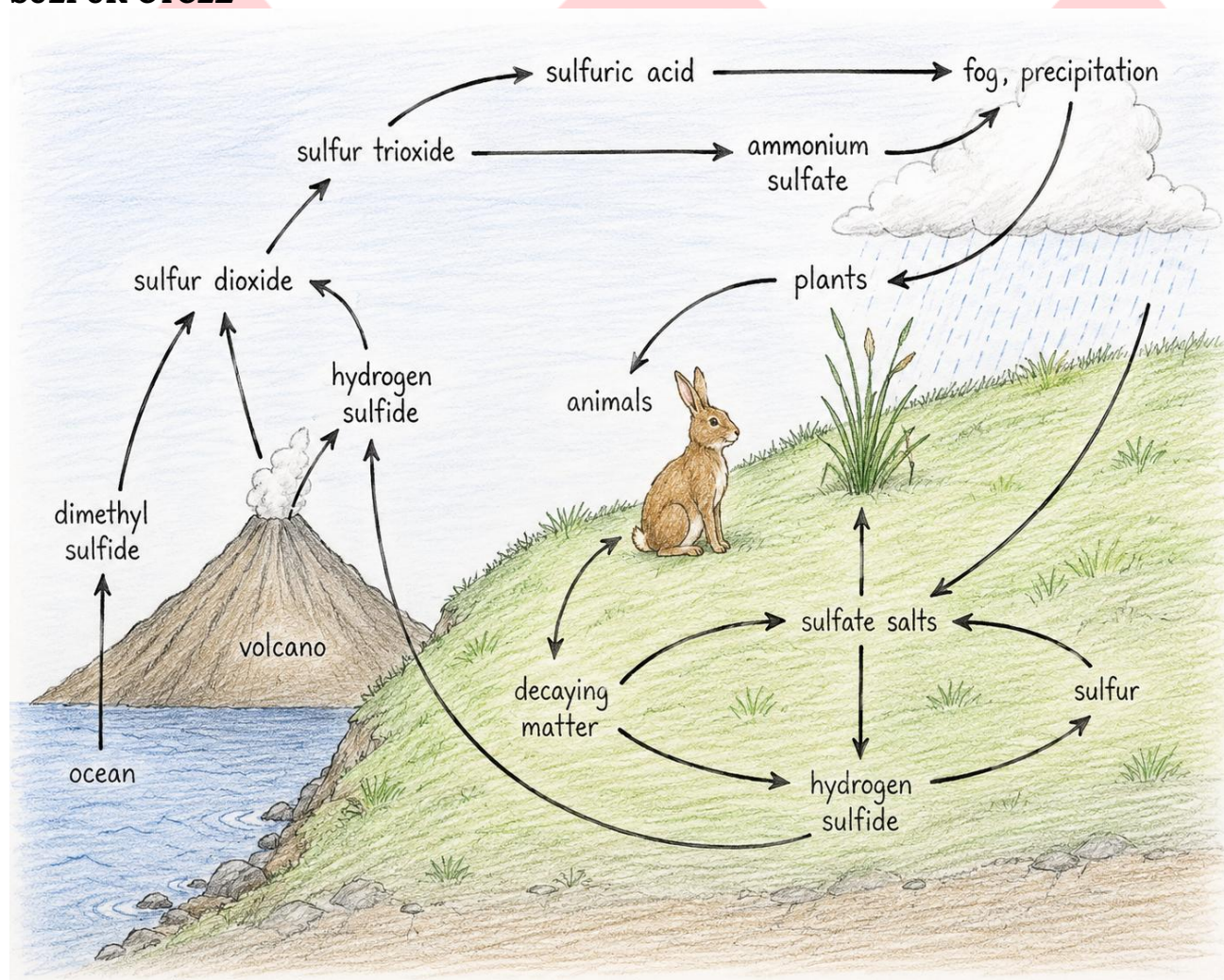


Phosphorus is an **essential nutrient** of both plants and animals. It is part of DNA molecules which carry **genetic information**. It is also part of **ATP** and **ADP**, which store chemical energy for use by organisms in cellular respiration and photosynthesis. Phosphorus is necessary for living organisms to make cell membranes, bones, teeth and shells.

There is **no gaseous form of phosphorus**; it is present in the atmosphere only as dust particles. Most phosphorus is stored in rocks and ocean sediments. Phosphorus has a simple cycle. When the reservoir of phosphorus rocks is exposed to water through the weathering condition, phosphorus can be released in the form of phosphate ions. When phosphate ions wash away, they soak into the soil; plants are starving for phosphate, so their roots absorb it.

Phosphates in the soil are taken into plants for lipid, nucleic acid and ATP synthesis and are passed through the food chains of ecosystems. When the animal dies, decomposing bacteria cycle the phosphate back into the soil, where it can be taken into plants or stay in the soil. If soil conditions are right, any phosphate remaining in the soil can slowly reform into rocks. Or washed out by rainfall into the drainage system which ultimately takes it to the sea. If it happens, it will be incorporated in marine sediments and so lost from the exchange pool. One important route for the rapid return of phosphorus from these sediments occurs where there is upwelling of ocean, bringing Phosphorus to the surface water, where it is taken into marine food chains. There is **no respiratory release** of phosphorus into the atmosphere.

SULFUR CYCLE

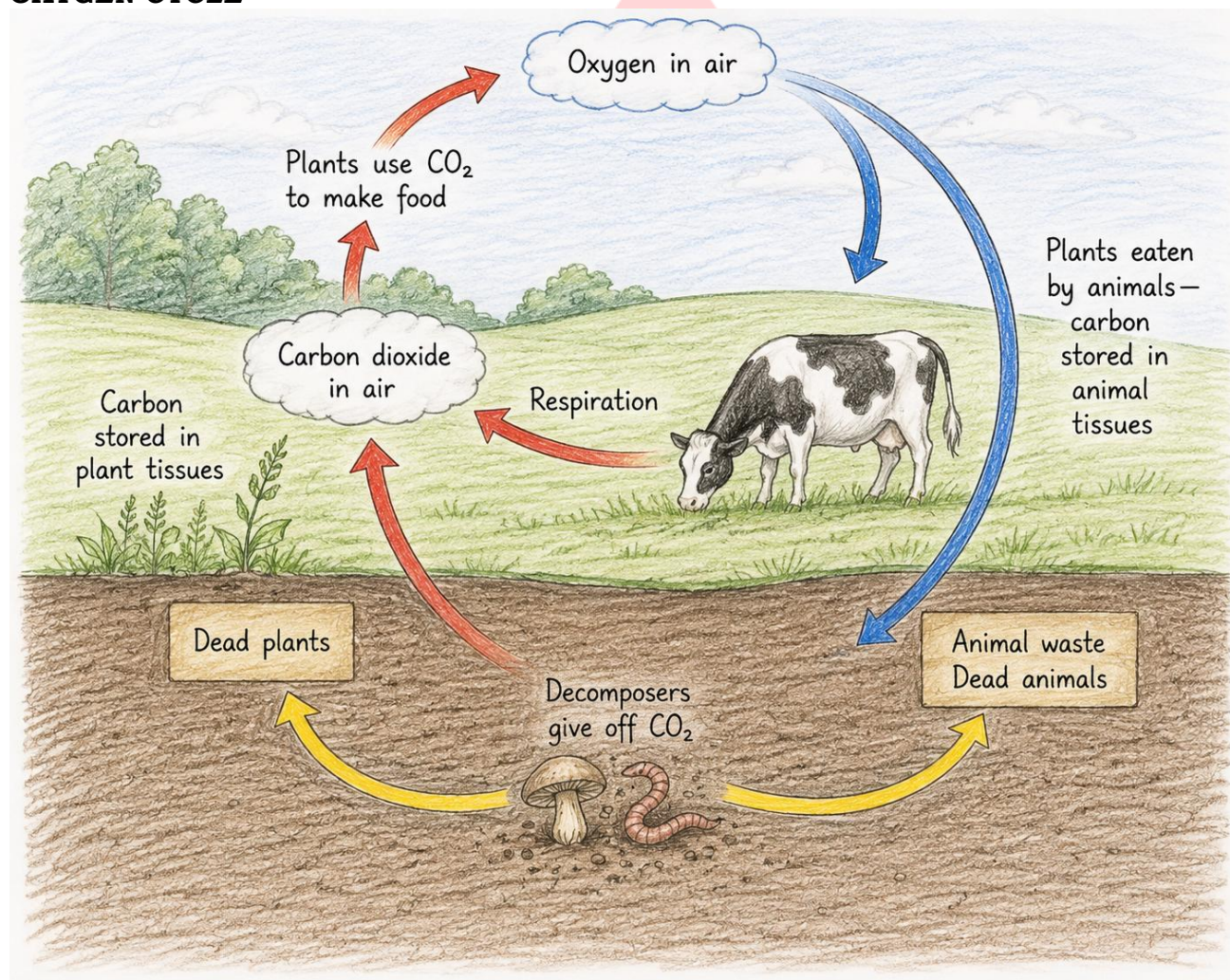


Many animals, including humans, depend on plants for sulfur-containing amino acids. Plants absorb Sulfur when it is dissolved in water. Animals consume these plants to take up enough Sulfur to maintain their health. Most of the Earth's Sulfur is tied up in rocks and salts or buried deep in the ocean in oceanic sediments. Sulfur can also be found in the atmosphere and enters it through both natural and human sources.

Source Type	How Sulfur Enters Atmosphere
Natural Sources	Volcanic eruptions, bacterial processes, evaporation from water, or decaying organisms.
Human Sources	Mainly industrial processes where Sulfur dioxide (SO ₂) and hydrogen sulphide (H ₂ S) gases are emitted on a wide scale.

When Sulfur dioxide enters the atmosphere, it will react with Oxygen to produce **Sulfur trioxide gas** (SO₃), or with other chemicals in the atmosphere to produce Sulfur salts. Sulfur dioxide may also react with water to produce **Sulfuric acid** (H₂SO₄). Sulphuric acid may also be produced from dimethyl sulfide, which is emitted to the atmosphere by plankton species. All these particles will settle back onto earth, or react with rain and fall back onto earth as **acid deposition**. The particles will then be absorbed by plants again and are released back into the soil, so the Sulphur cycle starts over again.

OXYGEN CYCLE



Oxygen is an important element for life on earth. Oxygen forms **21%** of the atmospheric air. The oxygen cycle is continuously used and released by different processes on earth. The oxygen cycle is **interconnected** with the **carbon cycle**. Producers (Green Plants and Phytoplankton) are the fixing members of atmospheric oxygen, they use CO₂ in the atmosphere to produce food and release O₂ as a by-product through the process of photosynthesis. The released O₂ by the producers is used by living organisms for their respiration, and the breakdown of carbohydrates into energy takes place and CO₂ is released to the atmosphere. This CO₂ is again used by producers. The cycle is complete and begins again.

The atmosphere is actually the smallest source of oxygen of the earth's total oxygen. The **largest**

reservoir of oxygen is the **lithosphere**, the Earth's crust. Most of this oxygen is not on its own or free-moving but part of chemical compounds such as **silicates** and **oxides**.

In the atmosphere, Oxygen is released by the process called **photolysis**. This is when high energy sunlight breaks apart oxygen-bearing molecules to produce free oxygen. One of the most well-known photolysis is the **ozone cycle**. O_2 oxygen molecule is broken down to atomic oxygen by ultra violet radiation of sunlight. This free oxygen then recombines with existing O_2 molecules to make O_3 or ozone. This cycle is important because it helps to shield the Earth from the majority of harmful ultra violet radiation turning it to harmless heat before it reaches the Earth's surface. The oxygen cycle is affected by clearing of vegetation, release of CFC (chlorofluorocarbon) gases and the effects of carbon cycle.

Importance of Biogeochemical Cycles

The biotic and abiotic components of the ecosystems are linked by flow of nutrients engineered by the biogeochemical cycles. When any of these cycles is affected, it leads to the unbalance of the natural world. Human activities, such as deforestation, fossil fuel burning, etc., have directly or indirectly affected the biogeochemical and physical processes. Altered biogeochemical cycles increase the vulnerability of biodiversity, food security, human health, air quality, water quality and changing climate. Managing and understanding environmental problems caused by climate change would require an understanding of the biogeochemical cycles.

Biogeochemical Cycles at a Glance

Cycle	Type	Main Reservoir	Key Process	Human Disruption
Water	Gaseous	Oceans (97.3%)	Evaporation, precipitation, infiltration	Deforestation, urbanisation, extraction
Carbon	Gaseous	Oceans (largest store)	Photosynthesis, respiration, decomposition	Fossil fuel burning, deforestation
Nitrogen	Gaseous	Atmosphere (78%)	Fixation, nitrification, ammonification, denitrification	Synthetic fertilizers, vehicle emissions
Phosphorus	Sedimentary	Rocks and ocean sediments	Weathering, uptake by plants, decomposition	Mining, agricultural runoff
Sulfur	Sedimentary	Rocks and oceanic sediments	Weathering, volcanic activity, bacterial processes	Industrial SO_2 emissions, acid rain
Oxygen	Gaseous	Lithosphere	Photosynthesis (releases O_2), respiration (consumes O_2)	CFC release, deforestation

PREVIOUS YEAR QUESTIONS (PRELIMS)

1. Most important anthropogenic source of both methane and nitrous oxide?
 - (a) Cotton
 - (b) Rice
 - (c) Sugarcane
 - (d) Wheat

2. In which biogeochemical cycle is weathering of rocks the main source?
 - (a) Carbon cycle
 - (b) Nitrogen cycle
 - (c) Phosphorus Cycle
 - (d) Sulphur Cycle

3. Which of the following are released into the atmosphere due to burning of crop or biomass residue?
 1. Carbon monoxide
 2. Methane
 3. Ozone
 4. Sulfur dioxide
 Which of the statements given above is/are correct?
 - (a) 1 and 2 only
 - (b) 2, 3 and 4 only
 - (c) 1 and 4 only
 - (d) 1, 2, 3 and 4

4. Consider the following statements :
 1. Agricultural soils release nitrogen oxides into the environment.
 2. cattle release ammonia into environment
 3. The poultry industry releases reactive nitrogen into compounds into the environment.
 Which of the statements given above is/are correct?
 - (a) 1 and 3 only
 - (b) 2 and 3 only
 - (c) 2 only
 - (d) 1, 2 and 3

5. Which best describes 'carbon fertilization'?
 - (a) Increased plant growth due to increased concentration of carbon dioxide in the atmosphere
 - (b) Increased temperature of Earth due to increased concentration of carbon dioxide in the atmosphere
 - (c) Increased acidity of oceans as a result of increased concentration of carbon dioxide in the atmosphere
 - (d) Adaptation of all living beings on Earth to the climate change brought about by the increased concentration of carbon dioxide in the atmosphere

6. Which of the following adds/adds carbon dioxide to the carbon cycle on planet Earth?
 1. Volcanic Action
 2. Respiration
 3. Photosynthesis
 4. Decay of organic matter
 Which of the statements given above is/are correct?
 - (a) 1 and 3 only
 - (b) 2 only
 - (c) 1, 2 and 4 only
 - (d) 1, 2, 3 and 4

7. Which of the following adds/adds Nitrogen to the soil?

1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation

Select the correct answer using the codes given below:

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

8. If phytoplankton of an ocean is completely destroyed:

1. The ocean as a carbon sink would be adversely affected.
2. The food chains in the ocean would be adversely affected.
3. The density of ocean water would drastically decrease.

Select the correct answer using the codes given below:

- (a) 1 and 2 only
- (b) 2 only
- (c) 3 only
- (d) 1, 2 and 3

9. Which of the above add carbon dioxide to the carbon cycle on Earth?

1. Photosynthesis
2. Respiration
3. Decay of organic matter
4. Volcanic action

Select the correct answer using the codes given below:

- (a) 1 and 4 only
- (b) 2 and 3 only
- (c) 2, 3 and 4 only
- (d) 1, 2, 3 and 4

10. Human activities have increased CO_2 , but a lot does not remain in the lower atmosphere because of:

1. Its escape into the outer stratosphere.
2. The photosynthesis by phytoplankton in the oceans.
3. The trapping of air in the polar ice caps.

Select the correct answer using the codes given below:

- (a) 1 and 2 only
- (b) 2 only
- (c) 2 and 3 only
- (d) 3 only

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